Angelsta Risk Assessment Can Our LLM Do It?

Feasibility review of the 26-component methodology, difficulty ratings, and MENA data sources with links

BASED ON: DATA REQUIREMENTS HANDOFF, 11 NOV 2026 (9 PAGES)

Verdict in one line: raw metrics YES, MENA external data YES with care, production scores and Company Risk NO until nine approval items are closed.

Review date: 11 November 2026 | Paper-based feasibility review, not an empirical test of the LLM
Web access from the review environment was blocked: links were verified through search results, see Section 9.

SECTION 1

The verdict

Short answer: yes for the data and raw-metric layer, yes with care for MENA external data, and no for production scoring. The Angelsta handoff is a clean, well-bounded specification. Its own text says that a developer can build the raw-metric engine now but must not invent the scoring rules. Our review agrees, and adds the MENA-specific risks that decide whether an LLM can do this reliably in practice.

Layer	Can our LLM do it?	Difficulty	Why
Read company files and extract fields (FM, INT)	**YES**, with source-cell citation and human spot-check	Moderate (2 to 4)	Arabic and English documents, mixed currencies, management accounts. Quality of the company's data is the bottleneck, not the model.
Compute the 26 raw metrics	**YES**, but in code, not in the LLM	Easy (formulas are simple)	All 26 formulas are arithmetic. Risk is in inputs and edge cases (zero or negative denominators, sign conventions).
Fetch MENA external data (EXT): FX, country risk, industry growth	**YES**, with two-source rule	Easy for FX, moderate for country, hard for industry	Official central-bank and statistics-office data exist for the main markets. Sector market sizes are thin and providers disagree.
Map raw metrics to 0 to 100, weight dimensions, Company Risk, Low/Medium/High	**NO** (blocked)	Not a technical problem	Nine decisions are unapproved. The LLM may draft options for approval but must not decide or silently assume them.
Prove the scores predict write-offs and down rounds	**NOT YET**	Needs outcome history	The handoff defers backtesting until enough realized outcomes exist. MENA exits are still thin (MAGNiTT notes only two tech IPOs in 2025).

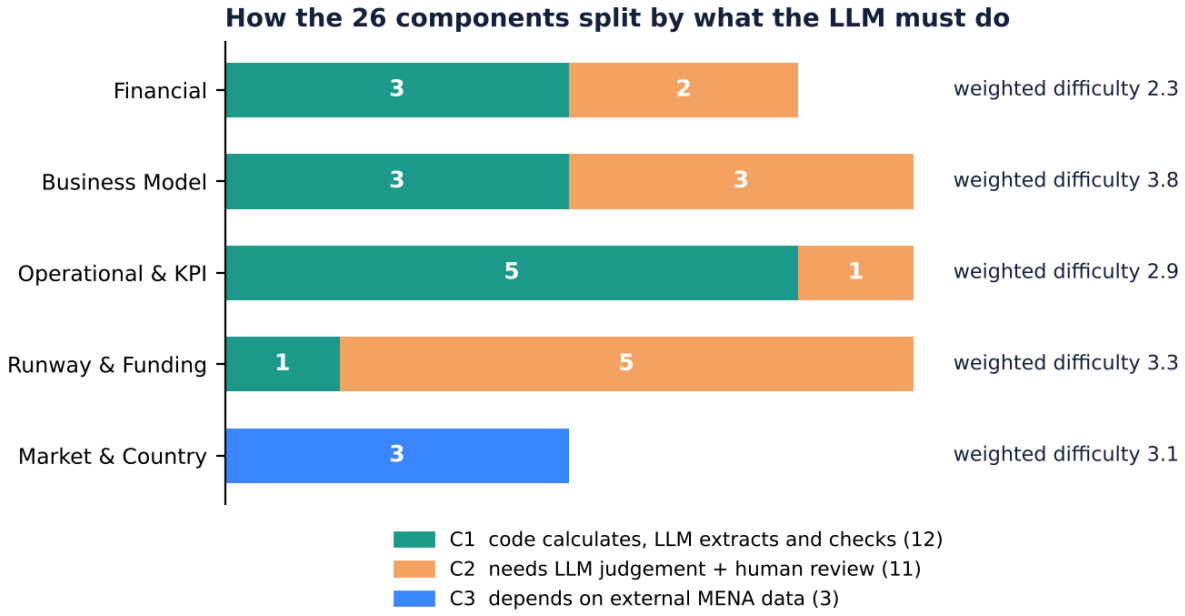


Figure 1. Of 26 components: 12 are code-computed with LLM extraction (C1), 11 need LLM judgement plus human review (C2), 3 depend on external MENA data (C3). Average difficulty is 3.1 out of 5. Ratings are this review's judgement, explained in Section 4.

What this review is, and is not

This is a **paper-based feasibility review** of the 9-page handoff. I did not test your LLM on real company files, and I did not have the 19-page source methodology. Section 7 gives the empirical test plan that turns this review into proof. Nothing here claims the LLM has been validated.

SECTION 2

What the handoff says, and what I checked

The handoff defines five business-risk dimensions and 26 components, each with an internal weight, a raw formula and a data package. It separates three kinds of evidence: **FM** (financial model), **INT** (company records) and **EXT** (independent industry, country and FX data).

Check	Result
Internal weights in each dimension sum to 100%	**Pass.** Financial 25+20+25+20+10; Business Model 15+20+15+25+15+10; Operational 25+20+20+15+10+10; Runway 30+10+15+20+15+10; Market 40+35+25. All equal 100.
Component count	**Pass.** 5 + 6 + 6 + 6 + 3 = 26 components under exactly 5 dimensions.
Every component has formula, data, source type	**Pass.** All 26 have a raw formula and a named evidence source (FM, INT or EXT).
Scoring rules are marked as unapproved	**Pass.** Section 7 of the handoff lists nine open decisions and forbids inventing them.
Market dimension needs external data only where stated	**Pass.** Only Industry Growth, Country Macro and FX use EXT. Sector Funding Momentum and Regulatory Risk are excluded in the MVP by design.
Source document available to me	**Limit.** I reviewed the 9-page handoff, not the 19-page source (v1.0, 8 Sep 2026). Any point decided in the source but omitted here may be missing from this review.

Observations for the methodology owner

These are gaps or ambiguities that matter once an LLM and developers implement the handoff. They are **observations, not errors** in the document: the handoff may resolve some in the full source. Each has a proposed way to close it, for your approval.

#	Observation	Why it matters for an LLM in MENA	Proposed resolution (for approval)
O1	Runway uses 'average monthly net burn' but the averaging window is not stated (R-A).	Different windows (3, 6, 12 months) give different runway for the same company. The LLM cannot choose.	Fix one window, show the window in output.
O2	'Accessible cash' is not defined (R-A, R-D).	Cash can be restricted, held in escrow, or hard to move out of a country. The Lebanese pound had multiple official rates for years; Egypt had official-versus-parallel gaps before its March 2024 float.	Define accessible cash and require bank confirmation; split restricted cash.
O3	'Total debt' is not defined (F-E).	Gulf and Egyptian startups use Islamic financing (murabaha, ijara, sukuk), shareholder loans, convertible notes and SAFEs. The ratio changes with each treatment.	Publish a debt-classification table, including Sharia-compliant financing and convertibles.
O4	Burn growth and KPI trend divide by the absolute previous value (R-B, O-B).	If the previous period is zero or the company turned cash-flow positive, the metric is undefined or reverses meaning. Doc flags zero but not sign flips.	Add explicit rules for zero and sign change; flag as non-computable.
O5	Possible overlap between components: OCF margin (F-D) and net burn (R-A, R-B); revenue growth (F-A) and KPI trend (O-B).	The same underlying movement can be counted twice once weights are applied.	Review correlation when the five dimension weights are approved.
O6	Sector classification for Industry Growth is not specified (M-A).	Without a taxonomy (for example ISIC) and a geography rule, two analysts pick different sectors and different growth rates.	Adopt one taxonomy; define country versus MENA-wide market; define multi-country companies.
O7	Country risk provider and scale are open (M-B).	Scales differ: OECD 0 to 7, Credendo 1 to 7, World Bank governance 0 to 100, agency letters. Several MENA countries are unrated by some providers.	Approve a provider hierarchy and a fallback for unrated countries.
O8	FX Risk is a 12-month point-to-point change in local currency versus investor base currency (M-C).	A pegged currency shows about 0 change even if peg-break risk is high. Revenue and cost currency mismatch inside the company is not captured.	Document as a known limitation; consider adding peg and mismatch flags later (needs methodology approval).
O9	Pre-revenue and early-stage companies cannot supply many inputs (F-A, F-B, B-E, O-D).	At seed stage many metrics are non-computable. Without a coverage rule, scores rest on very few components.	Decide a minimum-coverage rule inside the missing-data policy (item 03 of the handoff).
O10	Nominal figures in high-inflation or devaluing economies (F-A, B-E).	Nominal revenue growth can look strong while real or dollar growth is weak.	Decide whether to also compute constant-currency growth.

#	Observation	Why it matters for an LLM in MENA	Proposed resolution (for approval)
O11	Predictive validity cannot be tested yet.	Few realized write-offs in MENA, and thin exits, limit any backtest.	Use proxy outcomes (bridge needed, runway breach, down round) and plan a prospective study.

SECTION 3

How I rated difficulty

Each component gets a **class** (what the LLM must do) and a **difficulty** from 1 to 5 (how hard it is to do reliably end to end for a MENA startup, with LLM, code and human check).

Class	Meaning	LLM job	Human job
C1	Raw formula is plain arithmetic on structured fields.	Extract fields with citations; validate periods, units, currency.	Spot-check; approve the data package.
C2	Needs interpretation of contracts, plans, definitions or evidence.	Propose classification and evidence; flag conflicts.	Review and sign off every judgement.
C3	Needs independent external data (EXT).	Retrieve with provider, date and convention; cross-check two sources.	Approve provider hierarchy and mapping.

Score	Meaning
1	Trivial with structured input.
2	Easy; small judgement on definitions.
3	Moderate; needs data reconciliation or definition choices.
4	Hard; depends on unstructured documents or unapproved definitions.
5	Very hard; expert judgement and data that startups rarely hold.

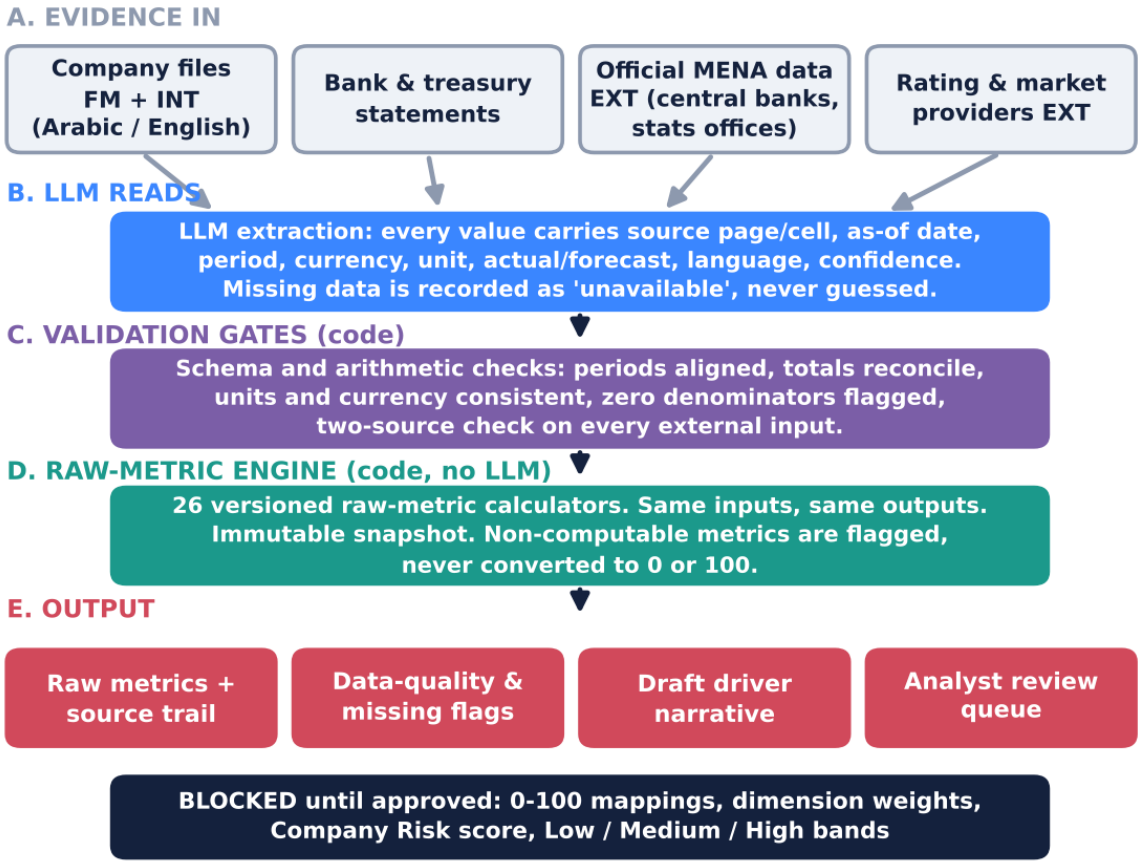


Figure 2. Reference pipeline. The LLM reads and drafts; code validates and calculates; scoring stays blocked until approved.

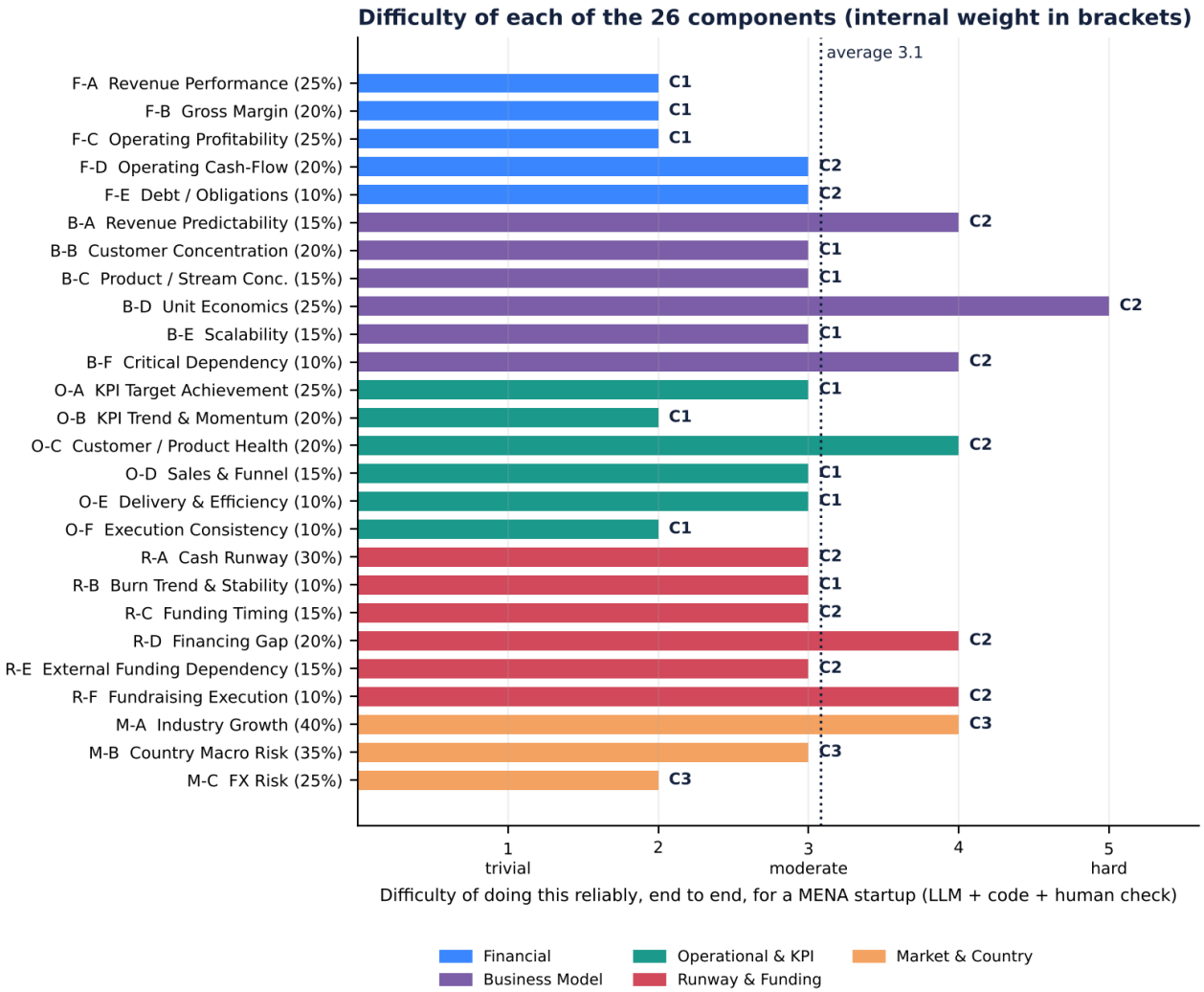


Figure 3. Difficulty by component. Business Model (weighted 3.8) is the hardest dimension; Financial (2.3) is the easiest. Unit Economics (B-D) is the single hardest component.

Where the LLM adds most, and least

- **Most value:** reading messy Arabic and English documents, reconciling them to a standard field list, and flagging inconsistencies. This is where analysts lose time.
- **Least value, highest risk:** doing arithmetic or choosing scoring thresholds. Both belong in versioned code and the approval process.
- **The real bottleneck is company data.** The handoff itself states that a financial model alone cannot supply all five dimensions.

SECTION 4

Component-by-component assessment

Source IDs such as **[S1]** point to the register in Section 9, which has full links. 'MENA watch-outs' are practical issues for startups in the region.

Dimension 1: Financial Risk (weighted difficulty 2.3)

ID	Component (weight)	Class / diff.	What the LLM does	MENA watch-outs	Sources
F-A	Revenue Performance (25%)	C1 / 2	Pull revenue for two comparable periods; check period length, currency, actual versus forecast.	Nominal growth misleads under inflation or devaluation (O10). Non-calendar fiscal years. Currency of reporting may differ from currency of billing.	Company FM; [S1] [S3] [S5] for FX translation
F-B	Gross Margin (20%)	C1 / 2	Extract revenue and COGS; map the company's cost lines to COGS.	Service and marketplace firms classify cost of revenue differently. Management accounts often lack a COGS line.	Company FM; peer benchmarks from listed firms [R-list]
F-C	Operating Profitability (25%)	C1 / 2	Extract operating profit or EBITDA (one only, per decision 04); list one-off items.	Add-backs for owner or related-party costs are common; must be documented.	Company FM
F-D	Operating Cash-Flow (20%)	C2 / 3	Extract cash-flow statement or rebuild from bank statements; reconcile to cash movement.	Many early-stage companies have no cash-flow statement. Rebuilding from Arabic bank statements needs verification.	Company FM + bank statements
F-E	Debt / Obligations (10%)	C2 / 3	Classify each liability as debt or not; sum; match to total assets on the same date.	Islamic financing, shareholder loans, convertibles and SAFEs need a rule (O3). Related-party balances.	Company FM; accounting standards [R-list]

Dimension 2: Business Model Risk (weighted difficulty 3.8)

ID	Component (weight)	Class / diff.	What the LLM does	MENA watch-outs	Sources
B-A	Revenue Predictability (15%)	C2 / 4	Read contracts; tag recurring or contracted revenue by template; list evidence.	Definition is pending per business-model template. Contracts may be in Arabic; government customers have long payment cycles.	Company contracts + FM
B-B	Customer Concentration (20%)	C1 / 3	Build customer revenue ledger; merge name variants; compute top-1 share and HHI.	Arabic and English spellings of the same customer must be merged. Customer names are personal data: pseudonymise before LLM use (Section 6).	Company ledger
B-C	Product / Stream Conc. (15%)	C1 / 3	Map revenue lines to approved streams; reconcile to total.	Stream taxonomy differs by company; needs template.	Company FM + INT
B-D	Unit Economics (25%)	C2 / 5	Compute contribution margin; gather CAC and LTV only where relevant, with definitions.	Hardest component: small samples, short histories, no common CAC/LTV definition. LLM must report 'not applicable' rather than estimate.	Company records
B-E	Scalability (15%)	C1 / 3	Compare change in operating cost to change in revenue over two periods.	Ratio is unstable with small revenue changes; two periods only.	Company FM
B-F	Critical Dependency (10%)	C2 / 4	Find channel, platform and supplier dependencies in contracts and cost ledgers; normalise to shares.	Normalisation basis is open. Dependence on a single platform, distributor or government buyer is common.	Company INT

Dimension 3: Operational & KPI Risk (weighted difficulty 2.9)

ID	Component (weight)	Class / diff.	What the LLM does	MENA watch-outs	Sources
O-A	KPI Target Achievement (25%)	C1 / 3	Read KPI dictionary; compute actual/target or target/actual by direction.	Needs approved targets and direction for each KPI; company templates differ.	Company INT
O-B	KPI Trend & Momentum (20%)	C1 / 2	Compute change versus previous period using |previous|.	Zero or negative previous values must be flagged (O4).	Company INT

ID	Component (weight)	Class / diff.	What the LLM does	MENA watch-outs	Sources
O-C	Customer / Product Health (20%)	C2 / 4	Select the KPI set for the template; extract retention, churn, usage, refunds.	Sub-weights pending. Definitions of 'active' vary widely.	Company INT
O-D	Sales & Funnel (15%)	C1 / 3	Extract CRM lead and opportunity counts; compute conversion and win rate.	Sales run outside a CRM are not visible. Stage definitions must be fixed.	Company CRM
O-E	Delivery & Efficiency (10%)	C1 / 3	Compute on-time rate, SLA achievement, defect rate where relevant.	Only where the model has such operations.	Company INT
O-F	Execution Consistency (10%)	C1 / 2	Count periods that missed target versus scored periods.	Short history: treat as confidence issue, not penalty (handoff rule).	Company INT

Dimension 4: Runway & Funding Risk (weighted difficulty 3.3)

ID	Component (weight)	Class / diff.	What the LLM does	MENA watch-outs	Sources
R-A	Cash Runway (30%)	C2 / 3	Read bank confirmation and cash forecast; compute accessible cash / average net burn; do not invent runway when net burn is zero or negative.	'Accessible cash' and averaging window undefined (O1, O2). Cash held in restricted or hard-to-move accounts.	Bank statements; company INT
R-B	Burn Trend & Stability (10%)	C1 / 3	Compute burn growth from a consistent series.	Needs same-definition burn each month; sign flips (O4).	Company INT
R-C	Funding Timing (15%)	C2 / 3	Derive months until funding need from the dated cash forecast.	Forecast quality varies; mark forecast versus actual.	Company FM + INT
R-D	Financing Gap (20%)	C2 / 4	Extract milestone, cash required and available resources; compute gap and gap ratio.	Milestone definition and 'available funding' need agreement with the company.	Company plan
R-E	External Funding Dependency (15%)	C2 / 3	Split funding need into internal and external portions.	Depends on honest classification of internal resources.	Company plan
R-F	Fundraising Execution (10%)	C2 / 4	Verify committed capital with documents; compute progress; combine with time remaining only after the joint rule is approved.	Soft circles versus signed commitments; SAFE or note terms. Joint rule is an open decision (item 07).	Term sheets, commitment letters

Dimension 5: Market & Country Risk (weighted difficulty 3.1)

ID	Component (weight)	Class / diff.	What the LLM does	MENA watch-outs	Sources
M-A	Industry Growth (40%)	C3 / 4	Map company to sector; fetch market size at two dated points or a published growth rate; record provider and date.	Sector market sizes in MENA are thin and providers disagree. Official value-added by activity is more reliable but coarser. 2026 shocks make any older series stale.	[S23] [S24] [S25] [S10] [S11] [S12]; MAGNiTT [S22] for venture context
M-B	Country Macro Risk (35%)	C3 / 3	Fetch rating or score from the approved provider; record scale, direction and date; map after approval.	Provider scales differ; some MENA countries unrated; ratings lag events. One aggregator showed conflicting Moody's entries for Egypt (Section 5).	[S16] [S17] [S18] [S19] [S14] [S20]
M-C	FX Risk (25%)	C3 / 2	Fetch spot and 12-month-prior rate from central bank; fix quote direction; compute change.	Pegs hide peg-break risk; Egypt floated on 6 Mar 2024; Lebanon used multiple rates; official versus parallel rates (O8).	[S1] [S2] [S3] [S5] [S7] [S9]

SECTION 5

External MENA data: what exists and how to use it

Only three components use external data, but they are where an LLM is most tempted to guess. The rule from the handoff is clear: **record provider and measurement date, and do not substitute unsourced estimates**.

5.1 FX Risk (M-C)

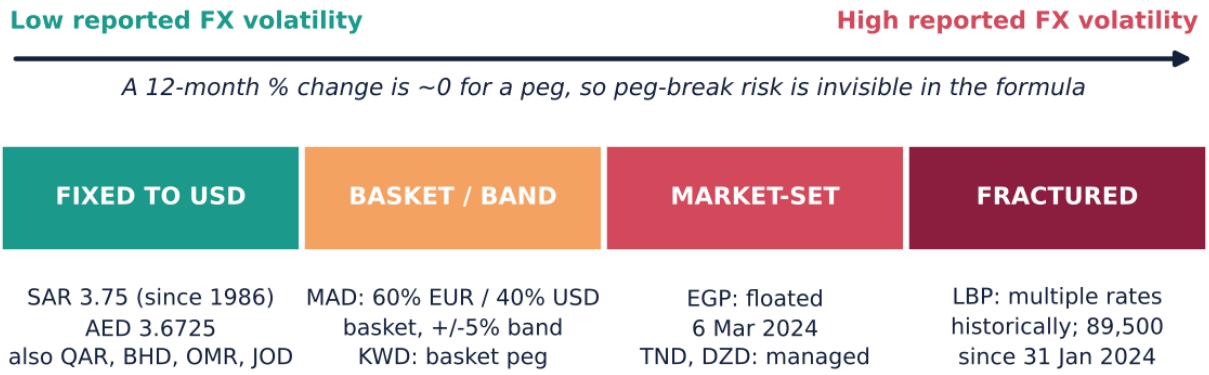


Figure 4. Currency regimes across MENA. Facts: SAR and AED pegs (SAMA, CBUAE); Morocco basket and band (Bank Al-Maghrib); Egypt float on 6 March 2024; Lebanon 89,500 per USD from 31 Jan 2024 (BDL-derived series). QAR, BHD, OMR and JOD pegs and the KWD basket are standard central-bank facts that I did not re-verify in this review.

Country	Currency and regime (verified item)	Where to fetch rates	Sources
Saudi Arabia	SAR fixed at 3.75 per USD since 1986 (SAMA currency page).	SAMA currency page and monthly bulletin.	[S3] [S4]
UAE	AED fixed at 3.6725 per USD (CBUAE monthly reference tables show 3.6725).	CBUAE monetary system and monthly FX files.	[S5] [S6]
Egypt	EGP floated 6 March 2024 with a 600 bp rate rise; press reported the pound lost over a third of its value that day. CBE page showed about 53.3 per USD on 10 Nov 2026 (search snapshot).	CBE market and official rates; daily interbank page.	[S1] [S2] [S31]
Morocco	Managed to a basket of 60% EUR and 40% USD; band widened to +/-5% on 9 March 2020. Check bkam.ma for any change since.	Bank Al-Maghrib band and central-rate pages.	[S7] [S8]
Lebanon	Official rate 89,500 per USD adopted in BDL balance sheet from 31 Jan 2024; multiple rates earlier. Most sources are secondary.	BDL (series mirrored by BLOM BRITE).	[S9]
Qatar, Bahrain, Oman, Jordan, Kuwait, Tunisia, Algeria, Iraq	Each central bank publishes its own official rate; pegs and regimes must be read from the central bank on the as-of date.	Central-bank root sites, Section 9 list R.	[R-list]

Rules for the LLM: (1) use the central bank's own published rate as primary; (2) fix the pair and quote direction before computing; (3) record the exact date of both observations; (4) if official and market rates differ, record both and flag it; (5) never take a free currency-converter page as primary. **Note:** the ECB reference-rate list does not cover most MENA currencies, so it cannot replace the local central bank.

5.2 Country Macro Risk (M-B)

Provider	Scale and coverage	Access	Use and caveat	Sources
S&P;, Moody's, Fitch	Letter ratings and outlooks for rated sovereigns.	Rating actions published by the agency; full history is a paid product.	Primary for rated countries. Many MENA countries are unrated by at least one agency.	[S20] (aggregator view)

Provider	Scale and coverage	Access	Use and caveat	Sources
Capital Intelligence (CI Ratings)	MENA-focused agency with a published sovereign ratings page and a 2026 publication calendar.	Sovereign page and calendar are public.	Useful regional second opinion; calendar shows when the next review is due.	[S17] [S18]
Credendo	Short-term and medium/long-term political risk on a 1 to 7 scale, plus business-environment grades.	Knowledge hub articles are public.	Shows 2026 conflict-related downgrades (for example Qatar short-term 4 to 5; Egypt 6 to 5; UAE business environment down). Re-read on the as-of date.	[S19]
OECD Country Risk Classification	0 (lowest risk) to 7; about 140 countries; set by an expert group for export-credit pricing.	Public table.	OECD states the scores are for setting export-credit minimum premiums and does not endorse other uses: treat as a cross-check only.	[S16]
World Bank Worldwide Governance Indicators	Six governance dimensions; 2026 method gives 0 to 100 scores; 200+ economies.	Public data and documentation.	Governance, not sovereign credit. Annual and lagged.	[S14] [S15]
IMF WEO / DataMapper; World Bank API	Macro series (growth, inflation, debt) by country, with API.	Free public APIs.	Good for context and for cross-checks; not a risk rating.	[S10] [S11] [S12] [S13]

Real example of why two sources are needed

On the Trading Economics Egypt rating page, the table showed Moody's at **B3, positive outlook**, while its own rating-news page listed Moody's as **Caa1, positive outlook, 7 March 2024** as the latest Moody's entry. The two pages disagree, so an LLM that read only one would silently carry an error. This is why the pipeline requires a second, primary source and flags conflicts instead of picking one.

5.3 Industry Growth (M-A)

- **Preferred primary data:** official value added by economic activity. Saudi Arabia's GASTAT publishes GDP by activity (its 2025 comprehensive update covers 134 activities versus 85 before) [S23]. Egypt's Ministry of Planning publishes quarterly GDP notes by activity based on CAPMAS data [S24], and CAPMAS publishes economic census metadata [S25]. IMF and World Bank APIs give cross-country series [S10] [S12].
- **Secondary data:** commercial market-size reports. Record publisher, date and method, and require a second source; they often disagree.
- **Venture context only:** MAGNiTT's FY2025 report puts MENA venture funding at about $3.8 billion over 688 deals, with Saudi Arabia $1.72 billion and the UAE $1.57 billion [S22]. Other outlets give slightly different totals, so use the primary report. This is funding activity, not market size, and the handoff excludes Sector Funding Momentum from the MVP.
- **Shock warning:** a GASTAT Q2 2026 release (as shown in search results) reports real GDP down 4.7% year on year with oil activities down 24.8% [S23]. Check the original PDF before using such figures. Sector series from before 2026 may no longer describe current conditions, so the as-of date must be shown.
- **Proposal for approval:** if the official-statistics growth and the commercial-report growth differ by more than a threshold set by Angelsta, flag as a source conflict and lower confidence (threshold not defined here).

SECTION 6

Company data realities and legal constraints in MENA

Internal data (FM and INT) is where most real-world failures will occur. The LLM must be built for these conditions from day one.

Issue	What happens in practice	Required control
Language and script	Contracts, invoices and bank statements in Arabic, English or both; Arabic-Indic digits; right-to-left tables in PDFs.	Test extraction on bilingual samples; normalise digits and decimal separators; keep original text beside the extracted value.
Currency	Reporting, billing and bank currencies differ; rates move (Egypt 2024, Lebanon).	Store original currency, rate used, rate date and source for every converted figure.
Accounting basis	IFRS, local standards, IFRS for SMEs, Islamic-finance standards for some firms, or unaudited management accounts.	Record the basis and audit status; do not mix bases between periods.
Fiscal calendars	Non-calendar fiscal years and different period ends.	Align periods explicitly (handoff acceptance check 01).
Actual versus forecast	Models mix actuals and plans in one sheet.	Tag every figure; the handoff requires this.
Informal evidence	Commitments and sales outside formal systems.	Count only documented evidence; list the rest as unverified.

Data-protection law decides where the LLM can run

Customer-level ledgers (needed for B-B) and contracts contain personal data. Three of the largest MENA markets have data-protection laws with cross-border transfer rules. **Summary below comes from law-firm commentary, not the legal texts. Have counsel confirm against the primary text before deployment.**

Country	Transfer rule (as reported)	Implication for hosting the LLM	Sources
Saudi Arabia (PDPL)	SDAIA transfer regulation under Article 29; transfers allowed under defined conditions; risk assessment required in some cases; SDAIA issued risk-assessment guidelines in early 2025. Revision dates differ between sources.	Prefer in-Kingdom processing or a documented transfer basis and risk assessment.	[S26] [S27]
Egypt (Law 151 of 2020)	Executive regulations issued late 2025; transfers abroad need a licence or an exception and a destination with comparable protection; the one-year compliance window ends around November 2026.	Treat the licence requirement as live; verify procedure with counsel.	[S28] [S29]
UAE (Federal Decree-Law 45 of 2021)	Adequacy-first approach with exceptions such as consent or contract; sector rules, for example banking data, can be stricter. Sources differ on the regulator's role.	Check current regulations and sector rules before sending data abroad.	[S30]

- **Minimise data:** send the LLM only the fields it needs. Replace customer names with IDs before processing (pseudonymisation).
- **Choose hosting by data class:** in-region or private deployment for personal data; public model APIs only for non-personal inputs.
- **Log everything:** prompts, model version, inputs, outputs, reviewer, correction reason (matches the handoff's output fields).

SECTION 7

How to prove it: the validation test plan

The handoff's acceptance checks are good, but they verify the engine, not the LLM. This plan tests the LLM's actual ability. **Pass thresholds below are my proposals**, to be agreed by Angelsta before testing starts.

#	Test	How to run it	Proposed pass gate
T1	Field extraction accuracy	Hand-label 200+ documents (financial models, contracts, bank statements) from at least six MENA countries, Arabic, English and bilingual. Compare each extracted field and its cited source.	At least 98% exact match on core numeric fields; 100% of values carry a valid source reference.
T2	Calculator correctness	Unit and property tests for all 26 formulas, including zero, negative and tiny denominators and reversed sign conventions.	100% pass; non-computable cases flagged, never converted to 0 or 100.
T3	Edge-case battery	Thousands versus millions; Arabic-Indic digits; mixed currencies; fiscal-year shifts; forecast labelled as actual; cash-flow-positive company.	No silent errors; every trap detected or flagged.
T4	Grounding and abstention	Remove inputs on purpose; ask the system to score anyway.	Zero invented values; 'unavailable' returned every time.
T5	External data reproduction	Recompute FX 12-month change for at least 10 MENA currencies from central-bank data; compare sovereign data across two providers; compare sector growth with official statistics.	FX matches to published precision; conflicts between sources are flagged, not hidden.
T6	Judgement agreement (C2 items)	Two analysts and the LLM independently classify predictable revenue, debt items, dependencies, KPI templates on 30 to 50 companies.	LLM-to-analyst agreement at least equal to analyst-to-analyst agreement; disagreements reviewed.
T7	Stability	Re-run the same companies with reworded prompts and different document orders.	Raw metrics identical; judgement items stable above an agreed rate.
T8	Privacy and security	Check pseudonymisation, hosting location, access logs, deletion.	Counsel sign-off for each country of operation.
T9	Predictive validity	Prospective study on proxy outcomes (bridge needed, runway breach, down round, write-off). Not possible until enough outcomes exist and scoring is approved.	Deferred, as the handoff states.

Readiness today

Area	Status	Next step
Raw-metric calculators (26)	Ready to build now	Implement with T2 and T3 tests.
Extraction from company files	Ready to pilot	Run T1 on a bilingual MENA sample.
External data retrieval	Ready to build; needs provider choice	Approve provider hierarchy (O7); run T5.
0 to 100 mappings, dimension weights, Company Risk, bands	**Blocked** by nine decisions	Decide; the LLM may draft options only.
Predictive claims	Not supportable yet	Plan a prospective study.

Minimum evidence record (align to the handoff, with MENA additions)

```
{ "company_id": "...", "as_of_date": "2026-11-11", "component": "R-A",
  "value": 13.3, "unit": "months", "status": "computed | non_computable | unavailable",
  "inputs": [ { "field": "cash", "value": 2400000, "currency": "USD",
                "source": "bank_statement.pdf p.2", "source_date": "2026-10-31",
                "basis": "actual", "language": "ar", "fx_rate_used": null, "fx_source": null } ],
  "methodology_version": "1.0", "confidence": "low | medium | high",
  "reviewer": "...", "correction_reason": null, "snapshot_ref": "immutable-id" }
```

SECTION 8

The nine decisions the LLM must not make

These are from the handoff, section 7. The MENA note says what is special for the region and what the LLM can safely prepare.

#	Decision	MENA note	What the LLM may prepare
01	Metric-to-100 mappings	Calibrate on MENA peer data, not only global benchmarks; listed-company data can help for margins.	Draft candidate curves with peer statistics for review.
02	Five dimension weights	Consider the effect of regional shocks on Market and Runway.	Run sensitivity tables on candidate weights.
03	Missing and stale data policy	Early-stage companies often lack data; a coverage rule is needed (O9).	Compute coverage statistics from pilot data.
04	Operating margin or EBITDA	Check what companies actually report.	Count how many pilot companies report each.
05	Within-component and template composites	Templates for SaaS, marketplace, e-commerce, services; MENA fintech and logistics variants.	Draft KPI dictionaries per template.
06	HHI, rating direction, FX quote conventions	Fix once, document, test (T5).	Show effect of each convention on pilot data.
07	Fundraising execution joint rule	Distinguish signed from soft commitments.	Propose scoring grids to review.
08	Confidence overlay	Combine completeness, freshness and source tier.	Propose a formula for approval.
09	Risk bands	Cut-offs need MENA outcome data.	Show distribution of raw outcomes; no bands without approval.

Recommended order of work

1) Close decisions 04 and 06 (small, fast). 2) Build the 26 calculators and tests T2, T3. 3) Pilot extraction (T1) on 30 to 50 MENA companies. 4) Choose country-risk and sector providers (O6, O7) and run T5. 5) Use pilot statistics to propose mappings and weights (decisions 01, 02, 03). 6) Approve scoring, then start a prospective validity study (T9).

SECTION 9

Source register and link verification

How links were verified, in plain words

The review environment could not open web pages directly (every request was blocked with HTTP 403). So **I could not click-test any link**. Instead each link below was checked by **appearing in a live web-search result on 11 November 2026** (status **V**), which confirms the page is published and indexed. Status **S** marks law-firm or press commentary: verify against the primary text. Status **R** marks an institution's root website that I know but did not open. Please click-test before external circulation.

ID	Source	Used for	St.	Link
S1	Central Bank of Egypt: exchange rates	EGP market rates (M-C)	V	https://www.cbe.org.eg/en/economic-research/statistics/exchange-rates
S2	Central Bank of Egypt: official rates	EGP official rates	V	https://www.cbe.org.eg/en/economic-research/statistics/cbe-exchange-rates
S3	Saudi Central Bank (SAMA): currency page	SAR peg	V	https://www.sama.gov.sa/en-US/FinExc/pages/currency.aspx
S4	SAMA: monthly bulletin statistics	SAR and banking data	V	https://www.sama.gov.sa/en-US/Statistics/Pages/MonthlyStatistics.aspx
S5	Central Bank of the UAE: monetary system	AED peg	V	https://centralbank.ae/en/our-operations/monetary-policy-and-domestic-markets/how-the-monetary-system-works/
S6	CBUAE: monthly FX reference file (July 2026)	AED reference rates	V	https://centralbank.ae/media/s3une40o/fx_jul26_en.pdf
S7	Bank Al-Maghrib: history of exchange-rate regime	MAD basket and band	V	https://www.bkam.ma/en/Find-information-about/Reform-of-the-exchange-rate-regime/History-of-the-exchange-rate-regime
S8	Bank Al-Maghrib: central rate and band calculation	MAD band	V	https://www.bkam.ma/en/Find-information-about/Reform-of-the-exchange-rate-regime/Calculation-of-the-central-rate-and-the-limit-rates-of-the-fluctuation-band
S9	BDL foreign-currency series (BLOM BRITE mirror)	LBP official rate (secondary)	V/S	https://brite.blominvestbank.com/series/BDL-Foreign-Currencies-11291
S10	IMF DataMapper API help	Macro data API	V	https://www.imf.org/external/datamapper/api/help
S11	IMF World Economic Outlook dataset	Macro series	V	https://data.imf.org/en/datasets/IMF.RES:WEO
S12	World Bank Indicators API documentation	Macro and sector API	V	https://datahelpdesk.worldbank.org/knowledgebase/articles/889392-about-the-indicators-api-documentation
S13	World Bank API basic call structures	API usage	V	https://datahelpdesk.worldbank.org/knowledgebase/articles/898581-api-basic-call-structures
S14	World Bank Worldwide Governance Indicators	Governance cross-check	V	https://www.worldbank.org/en/publication/worldwide-governance-indicators
S15	WGI documentation and methodology	WGI method	V	https://www.worldbank.org/en/publication/worldwide-governance-indicators/documentation
S16	OECD Country Risk Classification	0 to 7 cross-check	V	https://www.oecd.org/en/topics/country-risk-classification.html
S17	CI Ratings: sovereign ratings	MENA sovereign ratings	V	https://ciratings.com/regulatory-disclosures/sovereign-ratings/
S18	CI Ratings: sovereign ratings calendar 2026	Review dates	V	https://ciratings.com/wp-content/uploads/2025/12/Sovereign-Ratings-Calendar-2026-Issued-December-2025.pdf
S19	Credendo: business environment risk, Middle East shock	Political risk 1 to 7, 2026 moves	V	https://credendo.com/en/knowledge-hub/business-environment-risk-middle-east-shock-triggers-wave-downgrades
S20	Trading Economics: Egypt credit rating	Rating aggregator (needs primary check)	V	https://tradingeconomics.com/egypt/rating
S21	Trading Economics: credit rating API docs	Rating data via API	V	https://docs.tradingeconomics.com/indicators/credit-rating/
S22	MAGNiTT: FY2025 State of Venture Capital in MENA	Venture context	V	https://www.magnitt.com/research/FY-2025-MENA-VC-Report-51018
S23	Saudi General Authority for Statistics (GASTAT)	GDP by activity	V	https://www.stats.gov.sa
S24	Egypt Ministry of Planning: GDP and national accounts	GDP by activity	V	https://mped.gov.eg/Analytics?amp=&id;=61□=en

ID	Source	Used for	St.	Link
S25	CAPMAS: Economic Census 2022/2023 metadata	Sector structure	V	https://metadata.capmas.gov.eg/en/index.php/catalog/821
S26	Clyde & Co: Saudi risk assessment guidelines for transfers	KSA PDPL transfers	S	https://www.clydeco.com/en/insights/2025/03/update-on-saudi-arabia-risk-assessment-guidelines
S27	PwC: Saudi PDPL guide (Part 3)	KSA PDPL transfers	S	https://pwc.com/m1/en/media-centre/2024/documents/PDPL-Part-3-EN-2nd-Edition.pdf
S28	CMS: Egypt PDPL executive regulations	Egypt transfers	S	https://cms.law/en/are/legal-updates/egypt-s-pdpl-executive-regulations-issued-one-year-compliance-countdown-begins
S29	Kennedys: Egypt PDPL compliance countdown	Egypt transfers	S	https://www.kennedyslaw.com/en/thought-leadership/article/2026/egypt-s-personal-data-protection-law-the-compliance-countdown-has-begun
S30	Hunton: UAE federal data law	UAE transfers	S	https://huntonak.com/privacy-and-information-security-law/data-protection-in-the-uae-game-changing-federal-uae-data-law
S31	Daily News Egypt: CBE floats pound, raises rates 6%	Egypt float, 6 Mar 2024	S	https://www.dailynewsegypt.com/2024/03/06/egypt-floats-pound-raises-interest-rates-by-6-to-tackle-inflation/

R-list: institution root sites (status R, not opened in this review)

Institution	Use	Root link
Central Bank of Qatar	QAR rate	https://www.qcb.gov.qa
Central Bank of Bahrain	BHD rate	https://www.cbb.gov.bh
Central Bank of Oman	OMR rate	https://cbo.gov.om
Central Bank of Kuwait	KWD rate	https://www.cbk.gov.kw
Central Bank of Jordan	JOD rate	https://www.cbj.gov.jo
Banque du Liban	LBP rate	https://www.bdl.gov.lb
Central Bank of Tunisia	TND rate	https://www.bct.gov.tn
Bank of Algeria	DZD rate	https://www.bank-of-algeria.dz
Central Bank of Iraq	IQD rate	https://cbi.iq
Saudi Exchange	Listed-peer margins (benchmarks)	https://www.saudiexchange.sa
Abu Dhabi Securities Exchange	Listed-peer margins	https://www.adx.ae
Dubai Financial Market	Listed-peer margins	https://www.dfm.ae
Egyptian Exchange	Listed-peer margins	https://www.egx.com.eg
AAOIFI	Islamic-finance accounting standards	https://aaoifi.com
IFRS Foundation	Accounting standards	https://www.ifrs.org
Wamda	MENA startup and venture news	https://www.wamda.com

What I could not verify, stated plainly

- Direct page content: no page was opened, only search results were read.
- Current sovereign rating levels: not asserted anywhere in this report, because sources conflicted (Section 5.2). Fetch them on the assessment date.
- The Egypt rate near 53.3 per USD and Saudi Q2 2026 GDP figures come from search snapshots of official pages: confirm on the live page.
- QAR, BHD, OMR, JOD pegs and Kuwait's basket peg: standard central-bank facts from general knowledge, not re-verified here.
- Legal summaries (S26 to S30) are secondary commentary, not primary law.
- The 19-page source methodology and any real company data: not available to me.
- How your LLM performs: untested. Section 7 is the plan to find out.